

Urban–Hydrological Scenario and Global Comparison

Erasmus Mundus MUrCS Programme – Phase 1

Project: Enhancing Urban Resilience through Green Infrastructure and Climate-Smart Design

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1. Introduction

Nepal's urban centers, particularly the Kathmandu Valley, are facing increasing climate-induced flood risks due to rapid urbanization, altered rainfall patterns, and stressed drainage infrastructure. This master document integrates the national urban-hydrological scenario with a global comparison against a well-managed European city, fulfilling Erasmus Mundus Phase 1 requirements for contextual understanding and interdisciplinary validation.

2. Climate Change and Rainfall Dynamics in Nepal

Nepal is experiencing a shift toward High-Intensity Short-Duration (HISD) rainfall events. While total monsoon rainfall remains relatively stable, storm intensity has increased due to rising regional temperatures linked to global warming. For Kathmandu Valley, rainfall intensities up to 140 mm/hr (10-year return period) are now considered appropriate for climate-stressed urban flood modeling.

3. Urbanization and Groundwater Recharge Status

Unplanned urban expansion has increased impervious surface coverage beyond 85% in core areas of Kathmandu Valley. This has severely reduced natural infiltration and disrupted groundwater recharge. Despite being an ancient lake bed with high recharge potential in northern sandy zones, surface sealing has led to declining groundwater levels and increased flash flooding.

4. Roads, Drainage, and Infrastructure Stress

Most urban drainage systems in Nepal are combined sewer–stormwater networks that frequently exceed capacity during monsoon events. High sediment loads from construction activities block 50–70% of drainage capacity, forcing roads to act as temporary drainage channels and accelerating pavement degradation.

5. Green Infrastructure and Sustainable Urban Modernization

Green Infrastructure (GI) and Nature-Based Solutions (NBS) provide a sustainable alternative to purely grey infrastructure. Bio-swales, permeable pavements, and rain gardens can reduce runoff by 30–60%, with a conservative 40% reduction adopted for modeling. Infiltration rates of 2.5–4.0 cm/hr are appropriate for Kathmandu’s silt-dominated soils.

6. Comparative Urban–Hydrological Analysis: Nepal vs Europe

To contextualize Nepal’s challenges, Kathmandu Valley is compared with Copenhagen, Denmark, a globally recognized example of climate-resilient urban water management. Copenhagen’s proactive Cloudburst

Management Plan integrates green-blue infrastructure, separated drainage systems, and adaptive public spaces designed for extreme rainfall events.

Aspect	Kathmandu Valley (Nepal)	Copenhagen (Europe)
Urban Planning	Rapid, largely unplanned	Highly regulated, long-term
Imperviousness	>85% in core areas	Balanced with GI
Drainage System	Combined sewer-stormwater	Separated systems

Flood Design Level	Frequent exceedance (2–3 yrs)	100-year events
Green Infrastructure	Limited, pilot-scale	City-wide network
Ground Recharge	Declining	Actively enhanced
Sediment Control	High blockage (50–70%)	Low, well-managed

7. Conclusion

This integrated document strengthens the Erasmus Mundus MUrCS Phase 1 submission by combining a detailed national scenario with a global benchmark. It demonstrates the gap between current practice and achievable climate-resilient standards, while establishing a robust foundation for subsequent modeling, policy integration, and interdisciplinary collaboration.